

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE CODE: ITA 0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

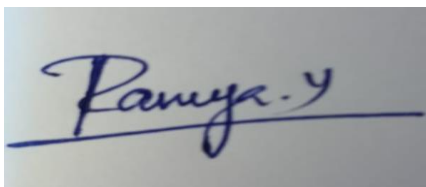
**Annexure A
CONCEPT MAPPING**

Code of Conduct:

I Ramya. Y/ (192424397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a white background. The signature is written in a cursive style and reads "Ramya. Y".

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.

Ans: Machine Learning System Blocks

A typical machine-learning system contains the following blocks:

- Data collection: Gather raw data from sensors, databases, web logs, or user inputs.
- Data preprocessing: Clean data, remove noise, handle missing values, and normalize features.
- Feature extraction/selection: Convert raw data into meaningful features used for learning.
- Model training: Use algorithms to learn patterns from training data.
- Validation and testing: Evaluate the model on unseen data.
- Prediction/decision making: Generate output such as classification or regression results.

Sketch

Raw Data → Preprocessing → Feature Extraction → Model Training → Validation/Test → Prediction

Examples

- Supervised learning: Email spam classification using labeled emails.
- Unsupervised learning: Customer segmentation using K-Means clustering.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.

Ans: Disaster Management Prediction System

- Data: Historical flood and earthquake records, rainfall, river level, seismic activity, and satellite data.
- Preprocessing: Handle missing values, normalize measurements, remove noise, and aggregate time-series data.
- Feature selection: Rainfall intensity, river flow, soil moisture, earthquake magnitude, depth, and location.
- Model choice: Random Forest for tabular risk prediction; LSTM for sequential time-series forecasting.

(a) Training, validation, and evaluation

- Split data into training, validation, and test sets.
- Train the model using historical labeled events.
- Tune hyperparameters using validation data.
- Evaluate with accuracy, precision, recall, F1-score, and RMSE for forecasting tasks.

(b) Assistance in disaster management

- Provides early warning alerts for high-risk regions.
- Supports evacuation planning and resource allocation.
- Helps authorities prioritize emergency response and reduce casualties.

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis.

Ans: Find-S Algorithm for Play Tennis

Positive examples: 3, 4, 5, and 7.

Example	Positive attributes	Hypothesis

Start	—	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
3	Overcast, Hot, High, Weak	$\langle \text{Overcast, Hot, High, Weak} \rangle$
4	Rainy, Mild, High, Weak	$\langle ?, ?, \text{High, Weak} \rangle$
5	Rainy, Cool, Normal, Weak	$\langle ?, ?, ?, \text{Weak} \rangle$
7	Overcast, Mild, Normal, Strong	$\langle ?, ?, ?, ? \rangle$

Final hypothesis: $\langle ?, ?, ?, ? \rangle$.

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

Ans: Candidate Elimination for Enjoy Sport

Example	S boundary	G boundary
Init	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$	$\langle ?, ?, ?, ? \rangle$
1 (+)	$\langle \text{Sunny, Warm, Normal, Weak} \rangle$	$\langle ?, ?, ?, ? \rangle$
2 (+)	$\langle \text{Sunny, Warm, ?, ?} \rangle$	$\langle ?, ?, ?, ? \rangle$
3 (−)	$\langle \text{Sunny, Warm, ?, ?} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?} \rangle \}$
4 (+)	$\langle \text{Sunny, ?, ?, Weak} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?} \rangle \}$
5 (−)	$\langle \text{Sunny, ?, ?, Weak} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?} \rangle \}$

Final boundaries: $S = \langle \text{Sunny, ?, ?, Weak} \rangle$, $G = \{ \langle \text{Sunny, ?, ?, ?} \rangle \}$.

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.

Ans: Find-S for Product Purchase

Positive examples: 3 and 4.

Example	Hypothesis
Start	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
3	$\langle \text{Middle, High, No, Fair} \rangle$
4	$\langle ?, ?, \text{No, Fair} \rangle$

Final hypothesis: $\langle ?, ?, \text{No, Fair} \rangle$.

6. In the context of the Find-S algorithm, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only.

Ans: Find-S for Workout

Positive examples: 2, 3, and 4.

Example	Hypothesis
Start	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
2	$\langle \text{Rainy, Mild, Evening, Happy} \rangle$
3	$\langle ?, \text{Mild, Evening, Happy} \rangle$
4	$\langle ?, ?, ?, \text{Happy} \rangle$

Final hypothesis: $\langle ?, ?, ?, \text{Happy} \rangle$.

7. In the context of the Find-S algorithm, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

Ans: Find-S for Loan Approval

Positive examples: 1, 2, and 4.

Example	Hypothesis
Start	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
1	$\langle \text{Employed, High, Good, Clean} \rangle$
2	$\langle \text{Employed, ?, Good, Clean} \rangle$
4	$\langle \text{Employed, ?, ?, Clean} \rangle$

Final hypothesis: $\langle \text{Employed, ?, ?, Clean} \rangle$.

8. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the specific (S) and general (G) boundaries. Show step-by-step updates after each training example.

Ans: Candidate Elimination for Subscription

Example	S boundary	G boundary
Init	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$	$\langle ?, ?, ?, ? \rangle$
1 (+)	$\langle \text{Young, High, Frequent, Mobile} \rangle$	$\langle ?, ?, ?, ? \rangle$
2 (+)	$\langle \text{Young, High, ?, Mobile} \rangle$	$\langle ?, ?, ?, ? \rangle$
3 (-)	$\langle \text{Young, High, ?, Mobile} \rangle$	$\{ \langle \text{Young, ?, ?, ?} \rangle, \langle ?, \text{High, ?, ?} \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$
4 (+)	$\langle \text{Young, ?, Frequent, Mobile} \rangle$	$\{ \langle \text{Young, ?, ?, ?} \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$
5 (-)	$\langle \text{Young, ?, Frequent, Mobile} \rangle$	$\{ \langle \text{Young, ?, ?, ?} \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$

Final boundaries: $S = \langle \text{Young, ?, Frequent, Mobile} \rangle$, $G = \{ \langle \text{Young, ?, ?, ?} \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$.

9. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the version space (S and G boundaries) and show the updates after each example.

Ans: Candidate Elimination for Healthy Lifestyle

Example	S boundary	G boundary
Init	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$	$\langle ?, ?, ?, ? \rangle$
1 (+)	$\langle \text{Balanced, Regular, Adequate, Low} \rangle$	$\langle ?, ?, ?, ? \rangle$
2 (+)	$\langle \text{Balanced, Regular, ?, ?} \rangle$	$\langle ?, ?, ?, ? \rangle$
3 (-)	$\langle \text{Balanced, Regular, ?, ?} \rangle$	$\{ \langle \text{Balanced, ?, ?, ?} \rangle, \langle ?, \text{Regular, ?, ?} \rangle \}$
4 (+)	$\langle \text{Balanced, ?, Adequate, Low} \rangle$	$\{ \langle \text{Balanced, ?, ?, ?} \rangle \}$
5 (-)	$\langle \text{Balanced, ?, Adequate, Low} \rangle$	$\{ \langle \text{Balanced, ?, ?, ?} \rangle \}$

Final boundaries: $S = \langle \text{Balanced, ?, Adequate, Low} \rangle$, $G = \{ \langle \text{Balanced, ?, ?, ?} \rangle \}$.

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

Ans: Health Analysis System

- Algorithms: Logistic Regression for interpretable disease prediction; Random Forest for handling nonlinear relationships and feature importance.
- Preprocessing: Handle missing values, remove outliers, normalize features such as blood pressure and glucose level.

- Feature selection: Use correlation analysis, mutual information, and feature importance scores.
- Training: Split data into training and test sets, train the model, tune hyperparameters, and evaluate using accuracy, precision, recall, and ROC-AUC.

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.

Ans: Machine Learning Pipeline

Data Collection → Preprocessing → Feature Engineering → Model Training → Evaluation → Deployment

- Data collection: Acquire raw data.
- Preprocessing: Clean and transform data.
- Feature engineering: Create useful input variables.
- Model training: Learn patterns from training data.
- Evaluation: Measure performance on validation/test data.
- Deployment: Use the trained model in real applications.

Training error vs. generalization error

- Training error: Error on the data used for training.
- Generalization error: Error on unseen data.
- Example: A model with 95% training accuracy but 75% test accuracy has poor generalization.

12. Design a machine learning model for spam email classification using labeled data.

(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).

(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.

(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

Ans: Spam Email Classification

(a) Choice of algorithm

- Naïve Bayes: Efficient for text classification and works well with word frequencies.
- SVM: Effective for high-dimensional text data.
- Logistic Regression: Provides probabilistic outputs and good interpretability.

(b) Preprocessing

- Tokenization: split emails into words.
- Stop-word removal: remove common words such as “the” and “is”.
- Stemming/Lemmatization: reduce words to root forms.
- Vectorization: convert text into TF-IDF or bag-of-words features.

(c) Evaluation metrics

- Accuracy: Overall correct classifications.
- Precision: Correctly identified spam among predicted spam.
- Recall: Correctly identified spam among actual spam.
- F1-score: Harmonic mean of precision and recall.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept Enjoy Sport.

(a) Explain the initialization of S (specific boundary) and G (general boundary).

(b) Show how S and G change with each training example.

(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

Ans: Candidate-Elimination with EnjoySport

(a) Initialization

- S boundary: Most specific hypothesis, initially $\langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$.
- G boundary: Most general hypothesis, initially $\langle ?, ?, ?, ?, ?, ? \rangle$.

(b) Updates

- For a positive example: generalize S minimally and remove inconsistent hypotheses from G.
- For a negative example: specialize G minimally and remove inconsistent hypotheses from S.

(c) Comparison with Find-S

Aspect	Find-S	Candidate-Elimination
Hypothesis	One most specific hypothesis	All consistent hypotheses
Negative examples	Ignored	Used to refine G
Generalization	Limited	Represents full version space

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities.

(a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).

(b) Explain feature scaling and handling missing data.

(c) Describe how overfitting can be detected and avoided.

Ans: House Price Prediction

(a) Choice of model

- Linear Regression: Simple and interpretable for linear relationships.
- Decision Tree: Captures nonlinear relationships and feature interactions.

- Random Forest: Often preferred because it reduces overfitting and handles complex data well.

(b) Preprocessing

- Feature scaling: Standardize numerical features such as size and number of rooms.
- Missing data: Use mean, median, or model-based imputation.

(c) Overfitting

- Detection: High training accuracy but poor test performance.
- Avoidance: Cross-validation, pruning, regularization, and limiting model complexity.

15. Explain the concept of clustering in unsupervised learning.

(a) Describe the working of the K-Means algorithm with a step-by-step example.

(b) Explain how the value of K is chosen using the Elbow Method.

(c) Compare K-Means with Hierarchical Clustering.

Ans: Clustering in Unsupervised Learning

Clustering groups similar data points without predefined labels.

(a) K-Means algorithm

- Choose K cluster centroids.
- Assign each data point to the nearest centroid.
- Recalculate centroids as the mean of assigned points.
- Repeat assignment and update steps until centroids stabilize.

Example

For $K = 2$, points are divided into two groups based on distance from the centroids; centroids are updated until clusters become stable.

(b) Elbow Method

Plot the within-cluster sum of squares (WCSS) against K. The optimal K is chosen at the “elbow” point where the reduction in WCSS starts to slow down.

(c) Comparison with Hierarchical Clustering

Aspect	K-Means	Hierarchical
Need for K	Required in advance	Not required initially
Output	Flat clusters	Dendrogram/tree structure
Scalability	Efficient for large datasets	More computationally expensive
Cluster shape	Best for spherical clusters	Can capture nested structures